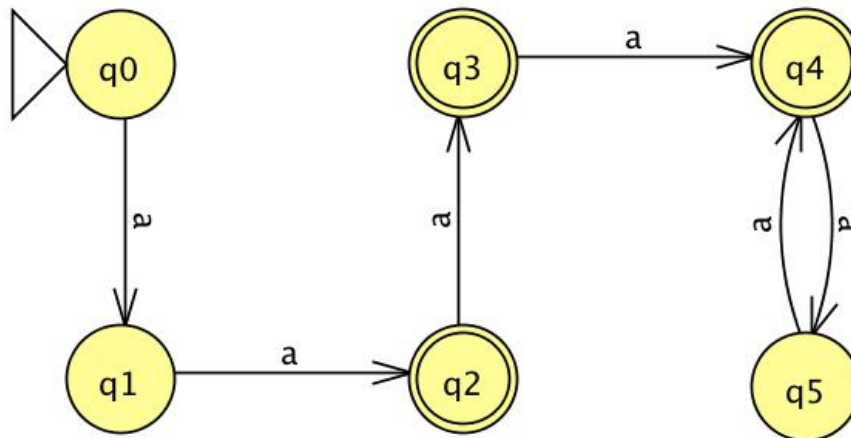


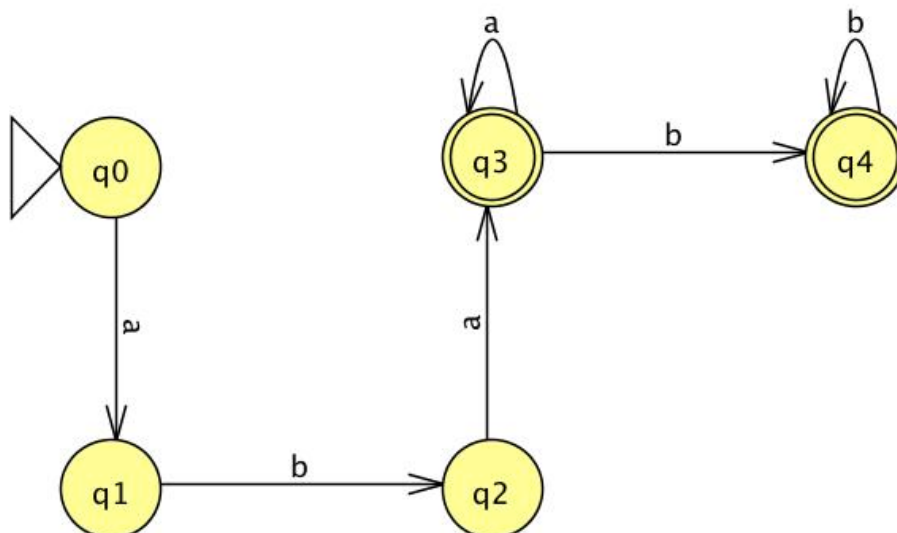
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pp. 54-56, Nos. 2, 12, 13, p. 62, Nos. 1, 3, pp. 68-69, Nos. 1,
For question (1) use the Pair Table method.

2) Find a DFA that accepts the language defined by the NFA in fig 2.8

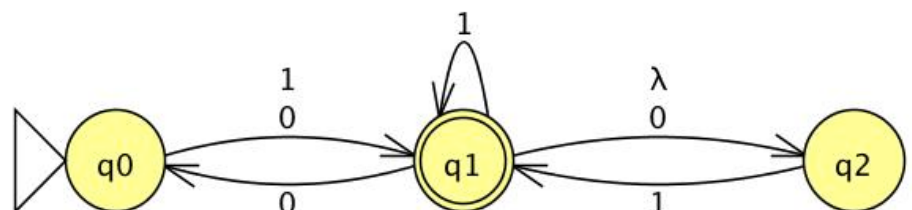


3) Design a NFA with no more than five states for the set $\{ abab^n : n \geq 0 \} \cup \{ aba^n : n \geq 0 \}$

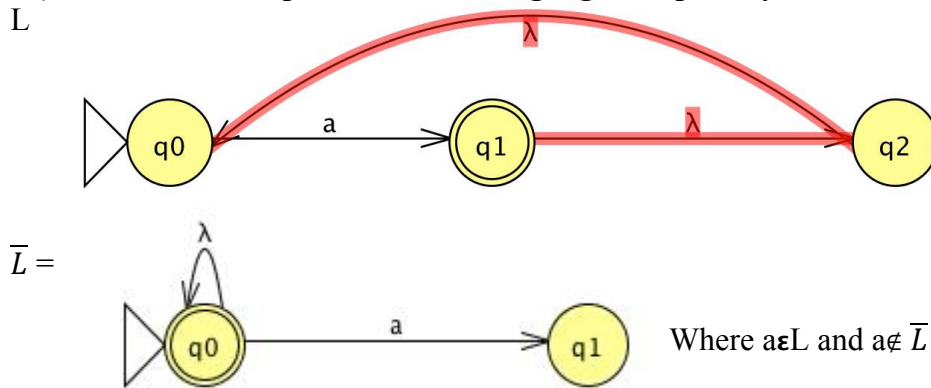


12) Which of the following strings 00, 010001, 10010, 000, 0000 are accepted by the following NFA.

Input	Output
00	Reject
010001	Accept
10010	Reject
000	Accept
0000	Reject

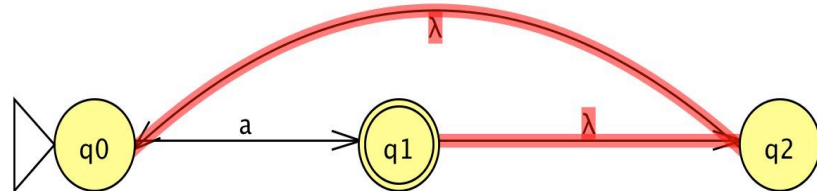


13) What is the complement of the language accepted by the NFA in fig 2.10?
L =



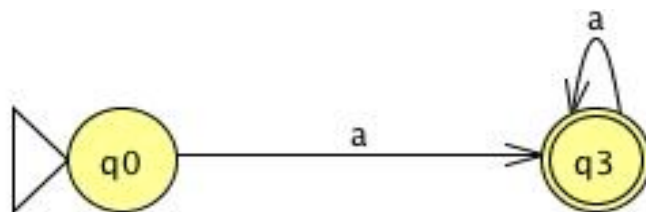
1) Use the construction from the theorem 2.2 to convert the NFA in fig 2.10 to a DFA.

NFA Transition Table		
Q	a	λ
q0	q1	-
q1	-	q2
q2	-	q0



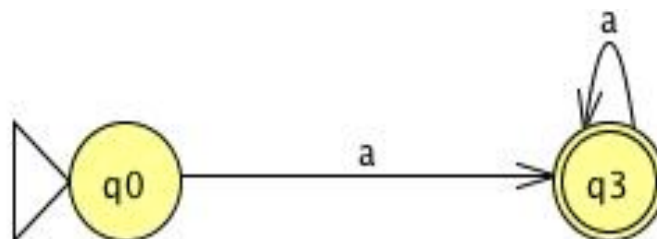
Transition Table without λ	
Q	a
q0	q1q2q0
q1	q1q2q0
q2	q1q2q0
q1q2q0	q1q2q0

$q0 \equiv q1 \equiv q2$, which means that $q0, q1$, and $q2$ are indistinguishable. Therefore, they are equivalent and we only have to choose one transition; which in this case is $q0$, as shown below.

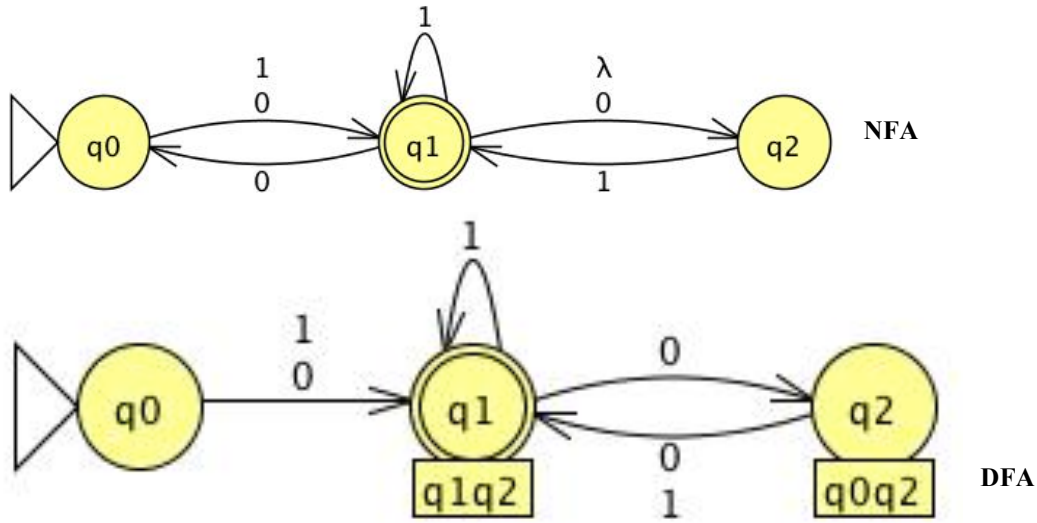


Can you see a simpler answer more directly?

Yes, it is quite obvious that the NFA is only accepting a , and looping on a . Therefore an optimum solution, from first sight, would be:

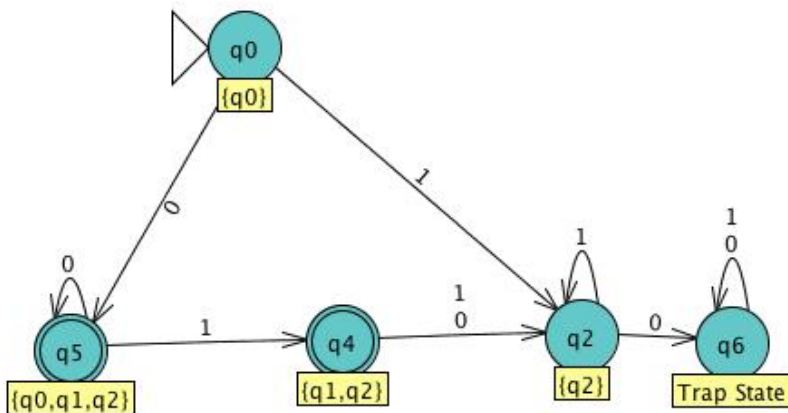
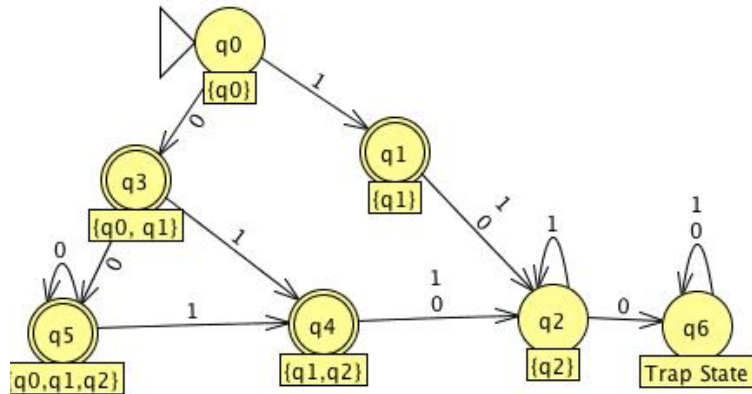


3) Convert the following nfa into AN equivalent DFA.



1) Minimize the number of states in the DFA in figure 2.16. Use the Pair Table Method

Pair Table Method		
Q	0	1
q0	q0q1	q1
q1	q2	q2
q2	$\emptyset$	q2
q0q1	q0q1q2	q1q2
q1q2	q2	q2
q0q1q2	q0q1q2	q1q2



Pair Table Method		
Q	0	1
q0	q0q1	q1
q1	q2	q2
q2	$\emptyset$	q2
q0q1	q0q1q2	q1q2
q1q2	q2	q2
q0q1q2	q0q1q2	q1q2

✓
✓

$q1 \equiv q1q2$, which means that $q1$, and $q1q2$ are indistinguishable. Therefore, they are equivalent and we only have to choose one

transition; which in this case is $q1q2$. The same happen for the states $q0q1$ and $q0q1q2$; since $q0q1 \equiv q0q1q2$, I only picked $q0q1q2$ as shown above.